

Thesis IV — Executive Summary

Spectral-Entropy Regularisation on Signed-Incidence Hypergraphs

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Question

Can a single regulariser, derived from the spectral entropy of a signed-incidence hypergraph view of a neural network, deliver *universal* improvement across architectures and datasets, or do the empirical positives reported in the literature collapse under matched cross-fixture testing?

Programme

18 phases, 2026-04-22 to 2026-04-29. Twelve regulariser *paths* (A–I plus combination arms) tested under a single benchmark harness on nine datasets (MNIST, FashionMNIST, KMNIST, CIFAR-10, EMNIST, SVHN, two_moons, spirals, circles) and five architectures (plain MLP, ResMLP, HighwayMLP, CapsMLP, plus the HyMeKo-emitted variants of the first three). Total volume: roughly 90+ paired-seed comparisons, ~3,000 individual training runs, 0 statistical-power failures, all CSVs reproducible from seeded scripts.

Settled positive — the universality claim

The normalised-entropy formulation

$$\mathcal{R}_B(A) = H_{\text{norm}}(A) = \frac{H(A)}{\log_2 \text{rank}(A)}$$

strictly Pareto-dominates the unnormalised form $\mathcal{R}_0 = \lambda H(A)$ across the universality matrix: every significant-positive effect of $\mathcal{R}_0$ is preserved or amplified, and *all four* significant-negative effects of $\mathcal{R}_0$ (capsule networks on MNIST, two-circle synthetic, deep ResMLP-20, Highway networks at depth) are neutralised ($p \geq 0.05$). The Kolmogorov–Arnold-derived target $H_{KA}^* = \log_2(2n+1)/\log_2 \text{rank}(A)$ supplies a principled per-dataset H^* and removes the one remaining free hyperparameter.

Settled negative — the activation-side family with a calibration law

The activation-side paths — `crosslayermi(PathF, pairwiseSanchez – GiraldoRényi – 2MI)` and

Networks emitted from `.hymeko` sources via the `torch_dataflow` backend (Tier-2 ResidualBlock / HighwayBlock blocks; `*hymeko` dataset entries in the bench harness) statistically mirror their hand-written PyTorch counterparts on the same regulariser arms, confirming that the canonical IR round-trips through to a runnable bench fixture without distorting regulariser dynamics.

Bounds (what is not yet shown)

Composability (Path B alongside dropout / batch-norm / optimiser weight decay) is queued as Phase 9; the three paper claims that hinge on it are not yet defended. *CapsMLP under activation-side regularisers* requires a refactor of CapsMLP's synthesised routing tensor before Phase 13 can be re-run. The `entropy_target_ka` ceiling clamp at 1.0 is a known implementation bug and should be 0.75 (matches the empirical H^* collapse threshold).

Artifacts

Code: `python/benches/thesis_iv_hard/` (harness, sweep scripts, aggregator). Generated nets: `python/benches/thesis_iv_hard/generated_nets/` + `data/nn/*.hymeko`. Reports: `reports/ph` (full taxonomy), `reports/ph12_path_i_negative_result.md` (negative-result writeup with the 13-run paired- Δ table), `reports/sanchez_giraldo_framework.{md,pdf}` (math for Paths E--H of the activation-side family), `RESULTS_VIEWS_SUITE.md` (aggregated all-experiments table). Every CSV under `data/benchmarks/` is seeded and reproducible by re-running the corresponding `ph*.sh` script.